

Checkpoint

An intelligent essentials dock that prevents you from forgetting important items before leaving home



Intelligent Essential-Management

Checkpoint is an intelligent essential-management system that helps users avoid forgetting everyday items like keys, wallets, phones, and identification cards. Rather than tracking lost items after they're misplaced, Checkpoint prevents the mistake before it happens.

The system combines physical storage, sensor-based detection, and smart suggestion logic to create a personal exit checkpoint that ensures users leave home with everything they need.



The Problem

Why We Forget

People frequently forget essential items when leaving home due to busy schedules, cognitive overload, inconsistent storage habits, and the challenge of remembering multiple small items.

Commonly Forgotten Items

- House keys
- Wallet
- Phone
- ID cards
- Access badges
- Earphones



Traditional Trackers vs. Checkpoint

Apple AirTag & Tile Trackers

Locate after loss - Help you find items after they're misplaced

Reactive solution - You've already made the mistake

Separate device needed - Requires attaching tags to each item

Checkpoint

Prevent before exit - Stops forgetting before it happens

Proactive system - Creates intelligent exit checkpoint

Centralized hub - Organized storage prevents disorganization

How Checkpoint Works

01

Item Removal Detection

When you pick up an item from the dock, sensors detect the action and check remaining items

02

Smart Suggestion

Mobile app suggests other essential items still in the dock

03

User Confirmation

Confirm or cancel suggestions through the app

04

Final Reminder

Subtle reminder before exit if items remain unconfirmed



System Architecture



Physical Docking Layer

Organized compartments with sensors for keys, wallet, phone, and accessories



Detection Layer

Weight, magnetic, or RFID sensors continuously monitor item presence



Processing Layer

ESP32 microcontroller manages detection logic and system communication



Application Layer

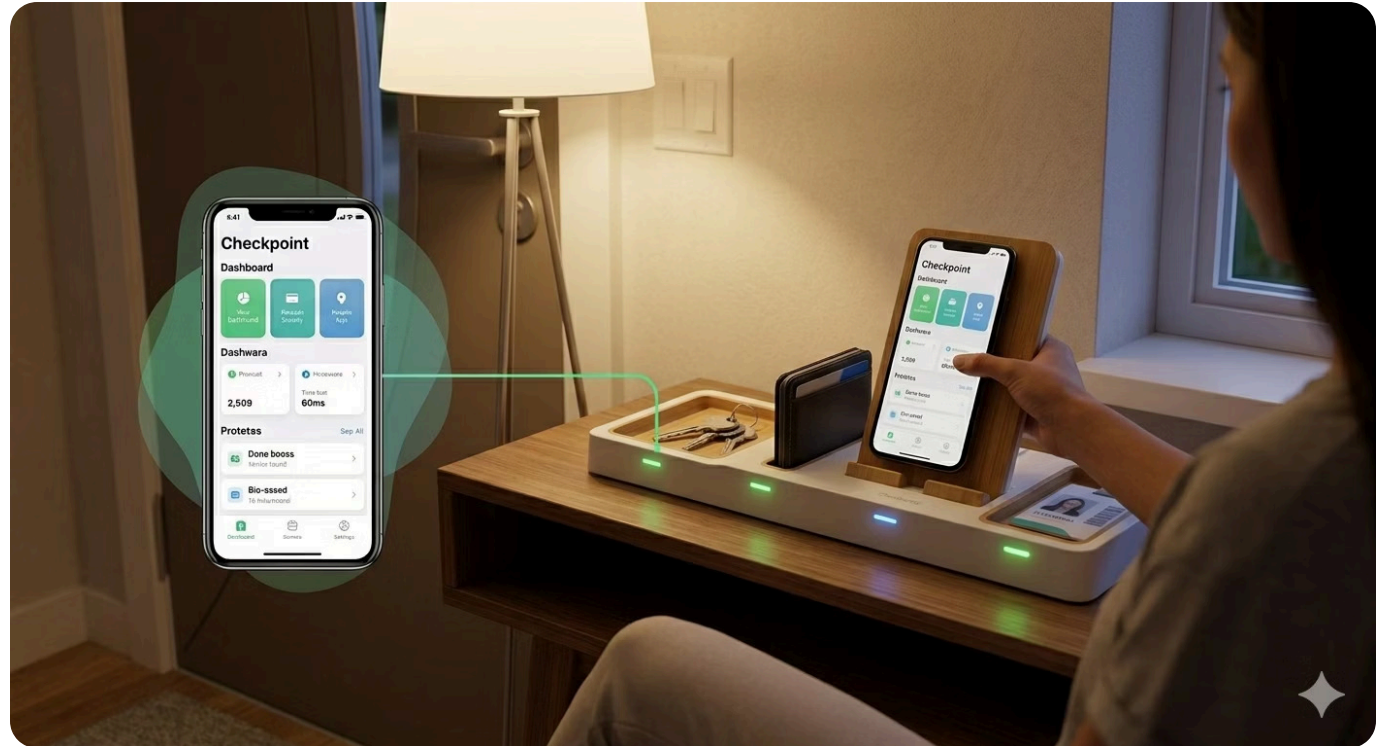
Mobile app provides user interface for configuration and confirmation

Intelligent Detection Mechanism

Behavior-Based Trigger

Instead of relying on door sensors or motion detection, Checkpoint uses **item removal as the trigger event**. When you pick up keys or phone, the system assumes you're preparing to leave.

This natural interaction creates seamless integration into your daily routine without additional steps or devices.



Returning Home

Place items in dock → Sensors detect placement → System updates status

Missing Items

App sends gentle reminder if expected items aren't detected

Flexible Dismissal

Dismiss reminders if item intentionally left behind

Visual Feedback System



Green LED

Item present and accounted for in the dock



Red LED

Item has been removed from the dock



Blue LED

Reminder active for unconfirmed items

Simple visual indicators allow users to quickly verify item status at a glance without opening the app.

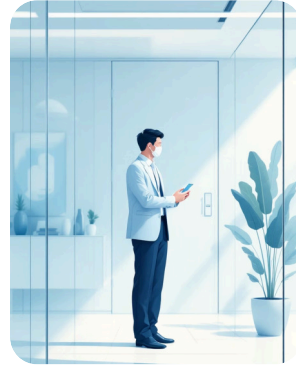


Real-World Use Cases



Students

Ensures ID cards, keys, and wallets are carried before leaving for classes



Working Professionals

Reminds users to carry office access cards, laptops, and business essentials



Families

Parents can configure essential items for children and monitor their preparedness



Shared Housing

Roommates can organize shared spaces for daily essentials with personalized configurations

Design Philosophy



Suggestion, Not Enforcement

Checkpoint focuses on **intelligent suggestion rather than intrusive alarms**. The system observes behavior, suggests missing items, and allows user confirmation.

- **Non-intrusive** - Gentle reminders rather than disruptive alarms
- **Adaptable** - Learns and adjusts to individual routines
- **User-friendly** - Natural interaction fits into existing habits